

Librika Platform User Manual

Functional Flow Specifications & API Call Flows

System Version:	v3.1.2 (LMS/ILMS Production Release)
AI Completions:	NVIDIA Nemotron completions API (30B parameter reasoning)
Security Level:	Role-Based Access Control (RBAC)
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Chapter 1: Functional Specifications & User Operations

This chapter outlines the functional behavior and operational flows of each user-facing subsystem within the Librika ecosystem.

A. AI Smart Book Scanner (Librarian Operations)

1. **Visual Capture:** The librarian targets a book cover. The camera feeds live image coordinates to the sequential frame processor.
2. **Parallel Scan Pipeline:** Librarians can snap and upload up to 10 books in parallel. The scanner returns recognized data in under 5 seconds.
3. **Stock Increment Logic:** If the scanned book is already cataloged, the database increments stock availability without creating duplicate records.

B. Gamified Reading Champion Leaderboard (Student Portal)

1. **Points Engine:** Evaluates student reading transactions. Awards positive points (+5 for checkouts, +15 for returning early, +20 for writing reviews, +50 for passing quizzes) and applies penalties (-20 for late returns, -50 for lost books).
2. **Dynamic Filters:** Students filter leaderboards by Class and Section, and view scores grouped by Weekly, Monthly, or All-Time activity.

C. AI Comprehension Quiz Module (Reading Validation)

1. **Minimum Reading Locks:** Quizzes remain locked based on book length (2 days for <100 pages, 5 days for 100-300 pages, 7 days for >300 pages) to prevent points exploits.
2. **Farming Cooldowns:** Applies a 90-day cooldown period on duplicate titles. Students cannot re-earn quiz points on the same book within 90 days.

D. AI Semantic Search Engine (Catalog Navigation)

1. **Concept Matching:** Users toggle AI Search to match queries by contextual meaning rather than literal strings (e.g. searching 'bravery' flags epic fantasies or historical biographies).
2. **Match Badging:** Renders gradient visual match indicators (e.g. ■ 85% Match) indicating context match probability.

E. Subscriptions & Billing Module (Admin Upgrades)

1. **Limits Validation:** Hard quotas restrict user and catalog additions based on plan levels (Free, Basic, Professional).
2. **3D Interactive Card Checkout:** Simulated modal flips 180 degrees when CVV fields focus, and updates card brand logos on prefix checks.

Chapter 2: Module API Call Flows & Technical Endpoints

Module A: AI Smart Book Scanner Call Flow

Step	Call Component / Endpoint	Action & Output Details
1	POST /admin/api/upload-cover	Takes captured camera frame. Resizes image under 600px width (for DB base64 compliance) and saves file. Returns static path URL.
2	POST /api/scan-vision	Passes the cover path. Encodes to base64, queries NVIDIA Nemotron Vision API to extract title, publisher, class, and ISBN. Strips LLM reasoning thought blocks before JSON load.
3	API Fallback 1: Google Books	If VLM returns empty critical fields, requests Google Volumes API for catalog data using title query (e.g. isbn:123).
4	API Fallback 2: DDG Web Search	If Google fails (NCERT/local books), queries DuckDuckGo for snippets. Passes crawled results to LLM for metadata correction.
5	POST /admin/api/save-scanned	Checks database. If unique ISBN/Title exists, calls /admin/api/add-copy to increment copies. Else, inserts new record and generates Code-128 barcode image.

Module B: Reader Leaderboard Call Flow

Step	Call Component / Endpoint	Action & Output Details
1	Issue Book (Physical/Digital)	Allocates +5 points to student score and inserts transactions record.
2	Return Book (Librarian return)	Evaluates returned date. If before due date: awards +15 points. If late: deducts -20 points. If lost: deducts -50 points.
3	Submit Review	Writes user review record. Once approved by admin moderation: awards +20 points.
4	AI Quiz Completion	Checks answers. If passed: awards +50 points to leaderboard totals.

Module C: AI Quiz Comprehension System Call Flow

Step	Call Component / Endpoint	Action & Output Details
1	GET /student/quiz//	Evaluates quiz eligibility parameters. Locks quiz if minimum reading period is not met based on page length (Short: 2d, Med: 5d, Long: 7d). Checks 90-day cooldown to block duplicate points farming.
2	ai_generate_quiz()	If quiz record does not exist in book_quizzes table, sends book metadata to NVIDIA completions API. Formats a 5-question multiple-choice JSON array.
3	POST /student/quiz//	Submits student answers. Grades attempt out of 5. Logs pass/fail metrics. If passed, awards +50 points to student profile and locks future attempts.

Module D: AI Semantic Search Engine Call Flow

Step	Call Component / Endpoint	Action & Output Details
1	GET /student/browse?q=[query]&ai;=true	User checks AI toggle and submits search. Intercepts request in Flask. Queries database for available catalogs.
2	perform_ai_semantic_search()	Constructs prompt containing query and book details. Passes parameters to completions endpoint to calculate relevance score (0-100) for each item ID.
3	Render student_browse.html	Filters items scoring 0. Sorts list by match percentage descending and renders a gradient ■ XX% Match badge on card overlays.

Module E: Subscriptions & Billing Portal Call Flow

Step	Call Component / Endpoint	Action & Output Details
1	Upgrade Select	Admin opens /billing. System reads limits and lists starter, basic, and professional packages.
2	Checkout Form Click	Pops up the 3D payment modal. Intercepts inputs to update card template. Flips 180 degrees on focusing CVV field.
3	POST /billing/checkout	Simulates transaction authorization, creates invoice logs, sets subscriptionStatus to 'active', and upgrades catalog quotas in SQLite.